

A Deep Learning Framework for Effective Anomaly Detection in Time Series Data

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Abstract— This Study presents a novel method for identifying anomalies in time series datasets by employing deep learning algorithms. Our approach blends autoencoders with Long Short-Term Memory (LSTM) networks. Although we concentrate our experiments on data related to electric vehicles (EVs), our overall objective is to show how flexible our method is for detecting anomalies in a variety of time-based datasets. The Autoencoder's capacity to rebuild input sequences and identify anomalies based on reconstruction mistakes is complementary to the LSTM network's ability to capture complex temporal patterns. By combining these methods, we provide a strong structure that may be employed to determine abnormalities in time series data and support well-informed decision-making in a variety of contexts. The accurate detection of abnormalities by our method improves the dependability of data analysis. Subsequent investigations may investigate additional improvements and expansions of LSTM and Autoencoder models with the goal of finding outliers in time series data.

Keywords— Time Series, Deep Learning Integration, LSTM-Autoencoder Approach, Temporal Pattern Recognition, Data Anomaly Identification, Electric Vehicles (EVs).

I. INTRODUCTION

In recent years, the exponential growth of data across diverse fields has emphasized the urgent necessity for robust anomaly detection methods, particularly within time series datasets. Anomalies, or outliers, signify data points that significantly deviate from expected patterns, often indicating critical events, faults, or irregularities within a system[1]. Detecting anomalies in time series data holds vital importance for applications spanning finance, cybersecurity, healthcare, and environmental monitoring. Traditional anomaly detection techniques like statistical methods and rule-based systems have shown limited efficacy, especially in dealing with complex temporal dynamics and high-dimensional data[1],[2]. As a result, new approaches to overcoming the difficulties involved in anomaly identification in time series data have become possible through the evolution of algorithms for deep learning. Deep learning algorithms give great opportunities for capturing complex temporal patterns and accurately identifying abnormalities due to their capacity to autonomously create hierarchical representations from data[3].

The objective of this research is to present a novel method for detecting anomalies in time series data by employing deep learning techniques, particularly Long Short-Term Memory (LSTM) networks and Autoencoders. Recurrent neural networks (RNNs) of the LSTM network type are excellent at representing temporal dynamics and long-range relationships,

which makes them a good choice for modelling sequential data like time series. [4]. Autoencoders, on the other hand, are neural network architectures trained to reconstruct input data and possess inherent capabilities to identify anomalies based on reconstruction errors. While our experimentation primarily centers on Electric Vehicles (EVs) data, [5] the overarching aim of our approach is to showcase its adaptability in identifying anomalies across various time-based datasets. By integrating LSTM networks and Autoencoders, we establish a robust framework for anomaly detection that not only enhances the reliability of data analysis but also fosters informed decision-making processes across diverse domains.

The following is the outline of the paper: A thorough overview of contemporary methods for anomaly identification in time series data is provided in Section 2. In Section 3, the methodology is explained in full, including the integration of LSTM networks and Autoencoders for anomaly detection. The experimental design, datasets utilised, assessment metrics applied, and experimental results are described in Section 4. Section 5 looks at the conclusions and ramifications of our methodology. The work is concluded in Section 6, which also suggests directions for future research.

II. LITERATURE REVIEW

There is a gap in time series anomaly identification, especially in IoT environments, as conventional anomaly detection approaches struggle to identify periodic and seasonal anomalies that are frequently seen in streaming data. In response to this problem, MOHSIN MUNIR et al. [6] presented DeepAnT, a novel deep learning-based method intended to efficiently identify a variety of anomalies in time series data. Using separate modules for time series prediction and anomaly identification, DeepAnT uses unlabelled data to anticipate typical behaviour. Because CNNs share parameters, it can train efficiently on short datasets and is appropriate for real-world applications with diverse sensor data. Evaluated against 15 algorithms on 10 benchmarks, DeepAnT consistently outperforms state-of-the-art methods. Long-term prediction poses a significant challenge in data mining, especially with Recurrent Neural Network methods, where accuracy diminishes rapidly with increased prediction steps. To address this, Xiang Yin et.al [7] presented VAECGAN, a framework comprising three components: an encoder net to encode exogenous sequences into latent space vectors, a generator net for data prediction, and a discriminator net for feedback to enhance prediction accuracy. Empirical studies on five real-world datasets (NASDAQ, SML, Energy, EEG, KDDCUP) validate the effectiveness and robustness of our approach, showcasing its potential to improve long-term

prediction accuracy in diverse applications. In complex systems with multivariate time series data, detecting anomalies is challenging due to intricate patterns and limited labelled data. Low accuracy is the outcome of existing approaches that lack regularisation or require large numbers of labels. Xuanhao Chen et al. [8] developed DAEMON, an unsupervised framework that makes use of adversarial autoencoder training, in order to overcome this. DAEMON achieves robust anomaly detection across various datasets by utilizing two discriminators and reconstruction error analysis. By using adversarial generation and hidden variable regularisation, it guarantees robustness. DAEMON also offers anomaly interpretation based on the reconstruction error of the constituent time series. DAEMON outperforms state-of-the-art techniques in experimental findings on real datasets, reaching a 0.94 F1-score and 97% anomaly interpretation accuracy.

Hardware technology advances emphasise how crucial it is to collect accurate Analyse time series data to identify abnormalities and mitigate risk, as noted by PEIHAI ZHAO et al. [9]. The diverse features, sizes, and interdependencies of sensor data pose obstacles in building a reliable multivariate time series anomaly detection system. A promising way to address these complications is through the use of deep learning techniques. In order to tackle this, we present MCRAAD, a novel technique to unsupervised deep learning designed to identify abnormalities in multivariate time series datasets. MCRAAD leverages a multilayer convolutional encoder to extract salient features and employs ConvLSTM units to capture temporal relationships effectively. Empirical findings on both simulated and actual datasets, including house monitoring, demonstrate MCRAAD's superior detection ability and robustness compared to baseline models, offering an effective solution for real-life multivariate time-series anomaly detection.

Dacheng Chen et al. [10] highlight the vulnerability of Cyber-Physical Systems (CPSs) to cyberattacks due to their complex and interconnected sensor networks. While continuous data streams from sensors enable intrusion monitoring, traditional detection algorithms struggle with the dynamic nature of CPSs. They provide GAN-AD, a novel method of anomaly detection for CPSs based on Generative Adversarial Networks. Using LSTM-RNN inside a GAN framework, they capture the multivariate time series distribution of sensors and actuators while taking potential interactions into account. Their technique surpasses prior methods with high detection rates and minimal false positives, effectively identifying abnormalities within a six-stage Secure Water Treatment (SWaT) system.

Sangkeum Lee et al. [11] created a revolutionary smart metering technique that incorporates anomaly detection for real-time residential power control systems. They identified outliers and missing values using smart metre data from 900 individual flat houses using a deep learning model known as the autoencoder. This model, combining a graph convolutional network and bidirectional long short-term memory network, provided efficient anomaly identification in smart meter data. They improved power management by merging a battery storage system and an electric car, as well as using multi-objective optimisation with the smart metering technique. Results demonstrated reduced electricity costs and grid dependency compared to power management without anomaly detection, showcasing the efficacy of the proposed

approach. Ruochen Jin et al. [12] The rapid rise in electric vehicles worldwide has led to increasing pressure on power battery disposal, necessitating efficient secondary use and recycling. To address this, we propose a framework for collecting battery charging data from various stakeholders, integrating an anomaly detection method based on Isolation Forest with two features. Additionally, a score-based mechanism is employed to screen data quality and ensure accuracy. Unlike previous approaches, our method leverages crowdsourced data to reduce sensing efforts and incorporates blockchain-based scoring to enhance data quality. Simulation results using NASA battery data indicate a significant improvement in F-measure criteria, up to 25.65%, compared to established anomaly detection algorithms. This approach offers promise for effective battery data collection and quality assessment, facilitating better estimation of battery state-of-health. Zafra-Mejía et al. [13] To better understand the behaviour of critical water quality measurements in the early components of a drinking water supply system (DWSS) in Bogota, Colombia, this study will employ transfer function ARIMA (TFARIMA) and autoregressive, integrated, and moving average (ARIMA) models. Eight years' worth of daily data on water quality were gathered from the drinking water treatment plant (WTP), reservoir, and river. The most important markers, as determined by a comparative examination of the DWSS components, were total iron, colour, and turbidity. In comparison to the river and reservoir, the WTP's turbidity behaviour was shown to be more influenced by historical data when the ARIMA analysis's time frame was extended. The TFARIMA models indicated that the impact of previous turbidity occurrences on the components was negligible. Data smoothing boosted the sensitivity of outlier identification, and ARIMA models were useful for researching

A novel method for identifying possible network assaults using the Auto-Regressive Integrated Moving Average (ARIMA) technique is presented by Su Fong Chien et al. [14]. This allows network administrators or automated management tools to take early preventive action. With the advent of new network paradigms like Software as a Service and Web Services, an early warning system is essential for maximising network utilisation. Their proposed solution harnesses predictive capabilities to provide feedback and facilitate timely intervention. Showing promising results, this technique extends beyond detecting Denial of Service (DoS) attacks, paving the way for automated defensive solutions with continued development.

A unique technique based on the Auto-Regressive Integrated Moving Average (ARIMA) model is introduced by Viacheslav Kozitsin et al. [15] for real-time diagnostics of complex technological systems, such power plants. The programme is meant to anticipate future system conditions and identify errors before they become serious, contributing to proactive maintenance and enhancing system reliability. This approach leverages the predictive power of ARIMA to support timely decision-making and optimize operational efficiency in critical technical environments.

It offers computational simplicity while providing effective solutions for anomaly detection and forecasting. Numerical experiments validate its performance across various scenarios. Additionally, the paper presents the ARIMAFD library, incorporating the proposed algorithms, suitable for real diagnostic systems. Overall, the developed

algorithm demonstrates efficiency and applicability in addressing anomaly detection and forecasting challenges in complex technical systems. Giulia Moschini et al. [16] employ the ARIMA model to examine the issue of detecting credit card fraud in unbalanced datasets without supervision. ARIMA predicts the typical expenditure patterns and identifies anomalies that arise from deviations or discrepancies. The study uses credit card datasets to test this approach's performance against four industry standard anomaly detection techniques: isolation forest, box plot, K-means, and local outlier factor. As can be seen from the results, the ARIMA model performs better than the benchmark models, indicating that it is a useful tool for identifying chargeback fraud. Using time series analysis to detect fraud in financial transactions is made easier by the insightful findings of this research.

III. METHODOLOGY

Through the integration of Long Short-Term Memory (LSTM) networks and Autoencoders, a novel mix of deep learning techniques, this study offers a groundbreaking way for identifying anomalies in time series data. Our methodology is designed to accurately capture complex temporal patterns and detect anomalies in time series information. Our approach provides a unique and efficient way to discover anomalies in time series analysis by combining the power of Autoencoders with LSTM networks. The main goal of the investigation is to show how the suggested approach may be tailored to work with various time-based datasets by using data from electric vehicles (EVs). Preprocessing is done on the dataset to deal with missing values, normalise features, and organise sequential input data so that deep learning models can be trained with it.

A. Model Architecture

The Figure 1. illustrates a ten-step process for training and testing a machine learning model. It begins with Data Extraction, where data is gathered from diverse sources. Following this, Data Cleaning is performed to eliminate duplicates, inconsistencies, and errors. Subsequently, Data Exploration is conducted to uncover patterns and trends within the cleaned data. Data Splitting divides the dataset into two sets: training and testing. Data processing begins, changing the data into a format that the model can understand, which may include scaling or converting variables. The following steps entail creating an Embedding, which is a mathematical representation of the data that is commonly utilised in activities such as natural language processing. Model training is the process by which a deep learning model learns patterns from training data. Model testing evaluates the model's performance by measuring accuracy against the testing data. The deep learning model is then applied to fresh data in Apply Software. Prediction Evolution evaluates the model's performance and adjusts it for greater accuracy. Additionally, the image mentions LSTM and Autoencoder, specific types of deep learning models useful in this process. LSTM is known for handling sequences, beneficial in time series data, while Auto encoder is adept at dimensionality reduction and feature learning. These models can be integrated into this flow for enhanced performance in training and testing machine learning models.

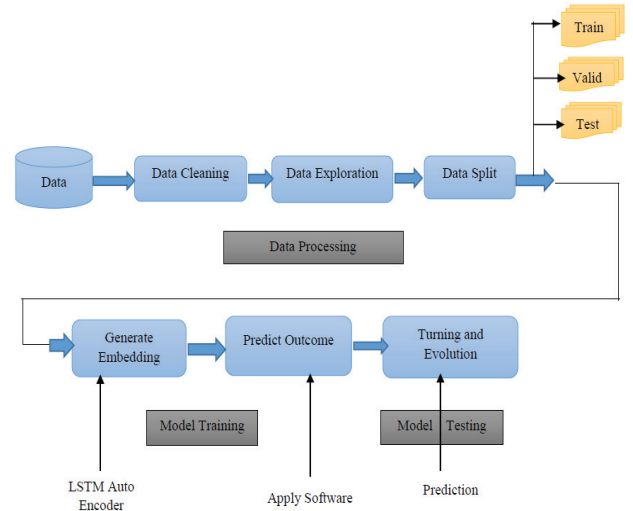


Fig. 1. System Architecture

The core of the methodology comprises two key components:

B. A Long Short – Term Memory (LSTM) :

LSTM networks are selected for their adeptness in modelling temporal dependencies and recording extensive contextual information over sequences of data. The LSTM architecture comprises multiple layers designed to learn intricate patterns and temporal dynamics inherent in time series data.

One key innovation of LSTMs is the incorporation of forget-gates and a sophisticated state flow mechanism, enabling the network to effectively learn long-term dependencies. The pivotal aspect of LSTMs lies in the presence of a linear self-loop that facilitates the flow of gradients without issues of exploding or vanishing gradients, which commonly afflict traditional RNNs. Despite the increased complexity of state computation in LSTMs, they exhibit substantial performance improvements. To elaborate on the computational process of LSTMs, we provide the equations governing the calculation flow, which are also illustrated in Figure 2. The input gate (denoted as i) and forget gate (denoted as f) are integral components of LSTMs, governing the extent to which the hidden state is influenced by both new inputs and the previous state at time step $t-1$.

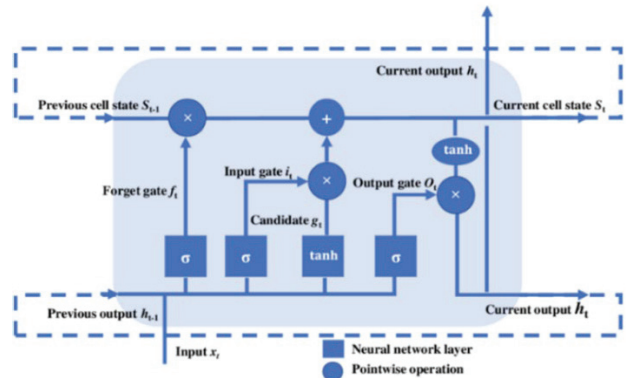


Fig. 2. Structure of a LSTM in a network

C. Autoencoders:

Autoencoders are important in anomaly detection because they recreate input sequences and assess the differences between original data and reconstructed equivalents. The encoder-decoder design of an Autoencoder is rigorously trained to minimise reconstruction loss, allowing the detection of anomalies characterised by substantial deviations during the reconstruction process.

An autoencoder is a form of neural network that learns a compressed or latent representation of training data, usually with fewer dimensions than the input. It has two main components: an encoder and a decoder.

The encoder converts input x into a hidden state h .

$$h = f(x) \quad (1)$$

and subsequently reconstructed into r via the

decoder $r = g(h)$

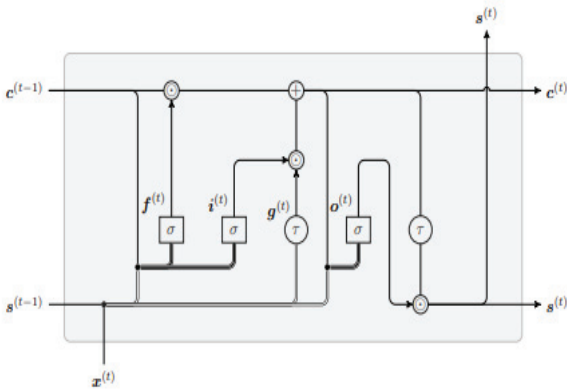


Fig. 3. Structure of a Autoencoders

Traditionally employed for feature learning and dimensionality reduction, autoencoders such as denoising variants excel in removing noise from corrupted input data by learning to predict the original, clean data. While standard autoencoders usually consist of a single layer for both the encoder and decoder, deeper architectures akin to feed-forward networks offer enhanced representational power and efficiency in learning complex functions.

Furthermore, autoencoders exhibit versatility in learning latent representations of sequential data through the integration of recurrent units within the encoder/decoder layers. In this configuration, the encoder maps an input sequence into a vector representation (h), with the dimensionality of this latent representation determined by the number of hidden units in the recurrent network. Within natural language processing, autoencoders find utility in sequence-to-sequence models, facilitating tasks like language translation and text summarization.

D. Training Process

1) LSTM Network Training:

The LSTM network is trained using historical sequences of time series data to learn the underlying patterns and dependencies. The network is optimized to predict future sequences and encode temporal information efficiently.

2) Autoencoder Training:

The Autoencoder is trained using a combination of normal (non-anomalous) sequences from the dataset. During training,

the Autoencoder learns to properly and efficiently rebuild input sequences, representing the data's underlying structure.

E. Anomaly Detection

The trained Autoencoder generates reconstruction errors, which are used to identify anomalies. Data points with significant differences between the original input and the reconstructed output are tagged as anomalous. The threshold for detecting anomalies can be determined using statistical analysis of the reconstruction errors or domain-specific knowledge. This model consists of an LSTM encoder, followed by an LSTM decoder. The LSTM Autoencoder model was trained on a dataset of time series sequences. The mean squared error (MSE) is utilised as the loss function in reconstruction. After training the LSTM Autoencoder, you may use the model to reconstruct input sequences and calculate reconstruction errors. Significant variances in reconstruction errors can help identify anomalies.

Residuals are the differences between observed (x) and predicted ($\hat{x}$) values from a model.

$$e = x - \hat{x} \quad (2)$$

Anomalies can be detected based on unusually large residuals that means outliers in the residual distribution.

Calculating Z-score:

The Z-score standardises a data point relative to the mean and the standard deviation. The Z-score normalises a data point based on the dataset's mean and standard deviation.

$$Z = (x - \mu) / \sigma \quad (3)$$

x is the data point of interest, μ is the mean, and σ represents the standard deviation. A Z-score indicates the number of standard deviations a data point is from the mean of the dataset. Anomalies can be identified based on a threshold Z-score ($|z| > 3$).

Anomaly detection in battery consumption in electric vehicles (EVs) is crucial for ensuring optimal performance and longevity of the battery systems. Two key methodologies for anomaly detection in this context are Long Short-Term Memory (LSTM) neural networks and autoencoders. For anomaly detection in battery consumption in EVs, a combination of LSTM and autoencoders can be used. LSTM can be employed to model the sequential nature of battery consumption data, capturing long-term dependencies. Autoencoders can then be used for feature learning or dimensionality reduction to identify anomalies in the learned representations. The combination of these techniques can provide a robust method for detecting anomalies in battery consumption in EVs, helping to improve battery performance and overall vehicle efficiency.

IV. RESULT AND DISCUSSION

A. Dataset

The dataset employed for anomaly detection in battery consumption within electric vehicles (EVs) encompasses detailed information about multiple vehicles. This includes their Vehicle Identification Number (VIN), geographical attributes such as county, city, state, and postal code, as well as vehicle specifications such as model year, make, model, electric vehicle type, and Clean Alternative Fuel Vehicle (CAFV) eligibility. Information on electric range, MSRP, legislative district, DOL vehicle ID, electric utility, vehicle

location coordinates, and the 2020 Census Trac are also included in the dataset. This comprehensive dataset facilitates the analysis of battery consumption patterns and anomalies in EVs across various parameters and geographical locations.

The dataset provides a comprehensive overview of various electric and hybrid vehicles, including their specifications and geographical details. It includes indicators denoting CAFV eligibility, electric range, and base MSRP. Additionally, geographical nuances such as legislative district and 2020 Census Tract are incorporated, along with unique identifiers like the DOL vehicle ID. The dataset serves as a valuable resource for analysing electric vehicle trends, pricing dynamics, and geographical dispersion.

B. Performance Metrics

- **Accuracy:** Accuracy is the fraction of correct predictions (TP + TN) out of total forecasts.

$$\text{Accuracy} = \frac{TP+NP}{TP+TN+FP+FN}$$

- **Precision:** The percentage of true positives among all positive forecasts.

$$\text{Precision} = \frac{TP}{TP+FP}$$

- **Recall:** Often referred to as sensitivity or true positive rate, represents the proportion of true positives out of all the real positives..

$$\text{Recall} = \frac{TP}{TP+FN}$$

- **F1 Score:** A balanced combination of accuracy and memory, or the harmonic mean..

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Utilizing the trained LSTM Autoencoder model, we can apply anomaly detection to time series data, with a specific focus on analysing electric vehicle (EV) battery consumption patterns. The system identified instances of abnormal battery usage, potentially indicating faults or irregularities in the vehicles' energy consumption patterns. We conducted a comparative analysis of our LSTM-Autoencoder model's performance against existing outlier identification techniques, such as Isolation Forest, IQR (Interquartile Range), and DBSCAN (Density-Based Spatial Clustering of Applications with Noise).

C. Comparison Methods

1) Isolation Forest:

Isolation Forest is particularly effective for anomaly detection in high-dimensional datasets because it does not rely on distance or density measures, which can be computationally expensive in such cases. It can handle large volumes of data efficiently by leveraging random partitioning and recursive subdivision to isolate anomalies effectively. This makes Isolation Forest a valuable tool for identifying outliers and anomalies in various types of datasets, including those with complex and diverse features.

2) IQR (Interquartile Range):

Dataset outliers are identified using the Interquartile Range (IQR) statistical approach. The value is obtained by subtracting the dataset's first and third quartiles. Outliers are data values below $Q1 - 1.5$ times the interquartile range (IQR) or above $Q3 + 1.5$ times them. The IQR technique is utilised

for outlier discovery in datasets when the distribution is not considered to be normal because to its resilience and simplicity.

3) DBSCAN (Density-Based Spatial Clustering of Applications with Noise):

The DBSCAN algorithm, which detects anomalies and clusters spatial data, is a robust density-based spatial clustering method utilised in the context of applications with noise. In contrast to conventional clustering approaches that necessitate predetermined cluster counts, DBSCAN discerns clusters autonomously by analysing the density of data points within the feature space. Dense regions of data points are delineated by regions of lesser density to constitute clusters. DBSCAN demonstrates exceptional performance when confronted with datasets containing irregular shapes and varying densities, thereby enabling its application in situations where alternative algorithms encounter difficulties. Furthermore, DBSCAN is capable of detecting anomalies, which are defined as data points that are neither cluster members nor located in low-density regions (noise).

TABLE I. THE PROVIDED TABLE COMPARES THE RESULTS OF THREE DIFFERENT OUTLIER DETECTION METHODS

Method	IQR	Isolation Forest	DBSCAN
Cleaned Shape	137669	158254	152493
Outlier Shape:	28766	8181	13942

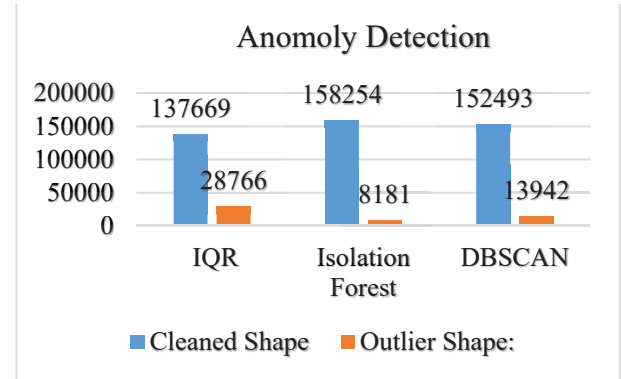


Fig. 4. Three outlier detection methods are compared in density-based spatial clustering of applications with noise (DBSCAN), IQR (Interquartile Range), and Isolation Forest)

These methods were employed to identify outliers within the 'Electric Range' and 'Base MSRP' columns of a dataset. The results indicate that the IQR method identified a substantial number of outliers, flagging 28,766 rows for removal. Consequently, the dataset was reduced to 137,669 rows after the removal of these outliers. In contrast, Isolation Forest detected a significantly lower number of outliers, marking 8,181 rows. As a result, the cleaned dataset retained 158,254 rows. DBSCAN, utilizing batching techniques, identified 13,942 outliers, resulting in a dataset containing 152,493 rows post-cleaning. Upon comparison, it is evident that Isolation Forest exhibited the least sensitivity to outliers, while IQR demonstrated the highest sensitivity by identifying the largest number of outliers. DBSCAN with batching fell between the two extremes, with a moderate number of outliers detected. Furthermore, the variations in the counts of outliers and the resultant dataset sizes underscore the distinct impacts of each method on the dataset. Achieving a balance between

outlier removal and the retention of meaningful data is crucial in selecting the appropriate outlier detection method. This decision should be guided by the objectives of the analysis and the acceptable level of data loss.

TABLE II. TRAINING LOSS PROGRESSION FOR LSTM MODEL

Epoch	Steps	Loss
1	67s 13ms /Step	0.00091946
2	66s 13ms /Step	5.0482E-06
3	63s 13ms /Step	3.2464E-06
4	65s 13ms /Step	2.2562E-06
5	63s 13ms /Step	1.5862E-06
6	65s 13ms /Step	1.2362E-06
7	63s 13ms /Step	8.6943E-07
8	62s 13ms /Step	7.6844E-07
9	65s 13ms /Step	6.9349E-07
10	64s 13ms /Step	5.1009E-07

TABLE III. AUTOENCODER TRAINING LOSS PROGRESSION

Epoch	Steps	Loss
1	14s 3ms/step	0.0094
2	13s 3ms/step	0.001
3	13s 3ms/step	0.001
4	13s 3ms/step	0.001
5	13s 3ms/step	0.001
6	13s 3ms/step	0.001
7	13s 3ms/step	0.001
8	13s 3ms/step	0.001
9	13s 3ms/step	0.001
10	13s 3ms/step	0.001

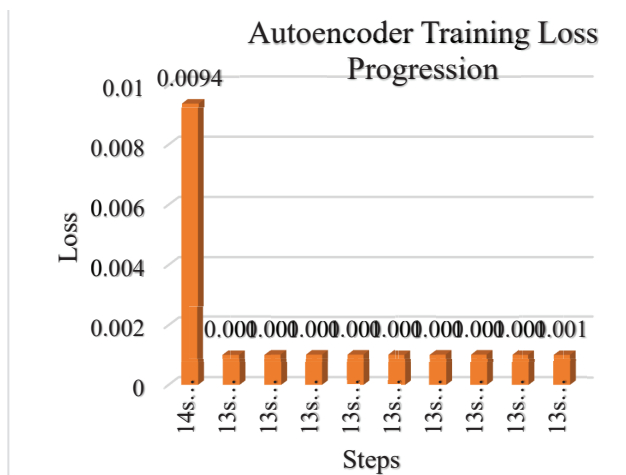


Fig. 5. Autoencoder Training Loss Progression

As shown in above figure 5. the progression of training loss for an autoencoder over 10 epochs. In each epoch, the time taken for training and the corresponding loss value are provided. Initially, the loss is relatively high at 0.0094, indicating significant reconstruction errors. However, as training progresses, the loss consistently decreases, reaching a minimum of 0.001 by the second epoch and maintaining this low value for subsequent epochs. The observed trend indicates that the autoencoder is successfully acquiring the ability to accurately reconstruct the input data with low discrepancy as the training progresses. The stability of the loss at 0.001 suggests convergence to a stable solution, indicating successful training of the autoencoder model.

TABLE IV. LSTM-AUTOENCODERS (PRECISION, RECALL, F1-SCORE AND ACCURACY)

Parameter	Value
Precision	0.071
Recall	1.0
F1- Score	0.133
Accuracy	92.851

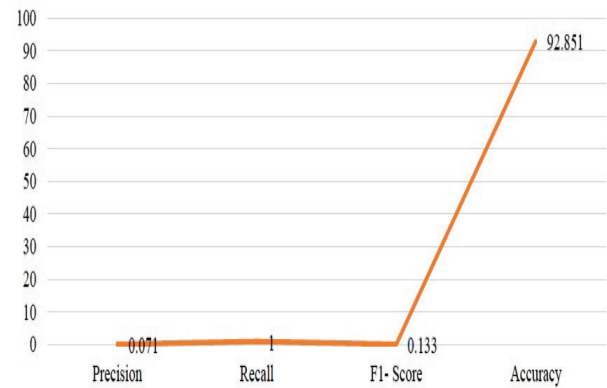


Fig. 6. performance curve

As shown in above figure 6. Density-Based Spatial Clustering of Applications with Noise analysis. Precision, measuring the accuracy of positive predictions, is notably low at 0.0837, suggesting a high rate of false positives. Recall, indicating the ability to find all positive instances, is at maximum, indicating the model captures all relevant instances but may also include noise. The F1-Score, which balances precision and recall, is relatively low at 0.1545. However, the accuracy of the model is high at 91.62%, signifying the overall correctness of predictions but with potential imbalances in precision and recall. Our findings highlight the trade-offs between precision, recall, and overall accuracy in anomaly detection using different methods. While our LSTM-Autoencoder demonstrated high recall but low precision, Isolation Forest and IQR showed varying degrees of precision and recall sensitivity. Moving forward, future research could focus on optimizing the LSTM-Autoencoder model parameters to improve precision without compromising recall. Additionally, investigating ensemble techniques or hybrid approaches combining multiple outlier detection methods could further enhance anomaly detection performance in EV battery consumption datasets.

V. CONCLUSION

The focus of study is to introduce an innovative deep learning-based methodology for detecting anomalies in electric vehicle (EV) battery consumption patterns. By integrating Long Short-Term Memory (LSTM) networks and Autoencoders, we achieve effective anomaly detection in EV energy consumption data. Our research addresses the critical need for robust anomaly detection in time series datasets, specifically focusing on managing the unique energy usage patterns of EVs. Leveraging advanced deep learning techniques, our approach contributes to enhancing anomaly detection capabilities within EV energy management. The LSTM-Autoencoder model demonstrates high recall but lower precision, effectively capturing anomalies with some potential for false positives. Additionally, traditional methods like Isolation Forest show promise in handling high-dimensional datasets. Moving forward, future research can explore hybrid approaches combining deep learning with statistical methods for improved anomaly detection. Optimizing model parameters and integrating contextual features will further enhance the performance and scalability of anomaly detection systems, providing actionable insights for optimizing EV operations and ensuring sustainable mobility.

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